

Problem

Several (at least two) nonzero numbers are written on a board. One may erase any two numbers, say a and b , and then write the numbers $a+2$ and $b-2$ instead. Prove that after any positive number of such operations, the set of numbers on the board cannot coincide with the initial set.

Source: St. Petersburg Mathematical Olympiad 1989

§1 The Critical Invariant

A natural quantity to monitor is the sum of the squares of all numbers currently written on the board. Let

$$S = \sum x_i^2.$$

We shall show that every allowed operation increases S strictly.

Suppose we choose two numbers a and b and replace them by $a+2$ and $b-2$. The contribution of these two entries to S changes from

$$a^2 + b^2$$

to

$$(a+2)^2 + (b-2)^2.$$

The increase is

$$\begin{aligned} (a+2)^2 + (b-2)^2 - (a^2 + b^2) &= 4a - 4b + 8 \\ &= 4(a - b + 2). \end{aligned}$$

Since the operation acts on a *set* of numbers, we may label the chosen pair so that $a > b$. Because the numbers are distinct elements of a set, we have

$$a - b > 0.$$

Consequently,

$$a - b + 2 > 2,$$

and therefore

$$4(a - b + 2) > 0.$$

Hence

$$(a+2)^2 + (b-2)^2 > a^2 + b^2.$$

It follows that the quantity S increases strictly after every operation.

§2 The Final Contradiction

The initial configuration has some fixed value S_0 .

After one operation the value of S becomes larger than S_0 , and after every subsequent operation it increases further. Thus, after any positive number of operations,

$$S > S_0.$$

If the resulting set of numbers were ever equal to the initial set, then the sum of their squares would also be equal to the original value S_0 . This contradicts the strict increase established above.

Therefore the set of numbers on the board can never coincide with the initial set after any positive number of allowed operations.

The initial set can never be recovered.